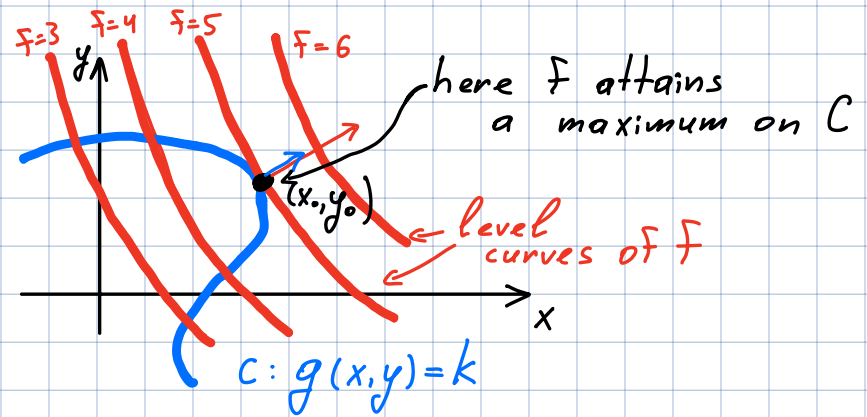


Last time:

Problem:

Find extreme values of $f(x,y)$
subject to constraint $g(x,y)=k$



level curves $g(x,y)=k$ and $f(x,y)=5$ touch at (x_0, y_0)

$$\Rightarrow \nabla f(x_0, y_0) = \underbrace{\lambda}_{\substack{\uparrow \text{a number, "Lagrange multiplier"}}} \nabla g(x_0, y_0)$$

$\uparrow$ a number, "Lagrange multiplier"

Method of Lagrange multipliers

To find maximum & minimum values of $f(x,y)$
subject to constrain $g(x,y)=k$:

(a) Find all values of x, y, λ such that

- $\nabla f(x,y) = \lambda \nabla g(x,y)$
- $g(x,y) = k$

$$\begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$$

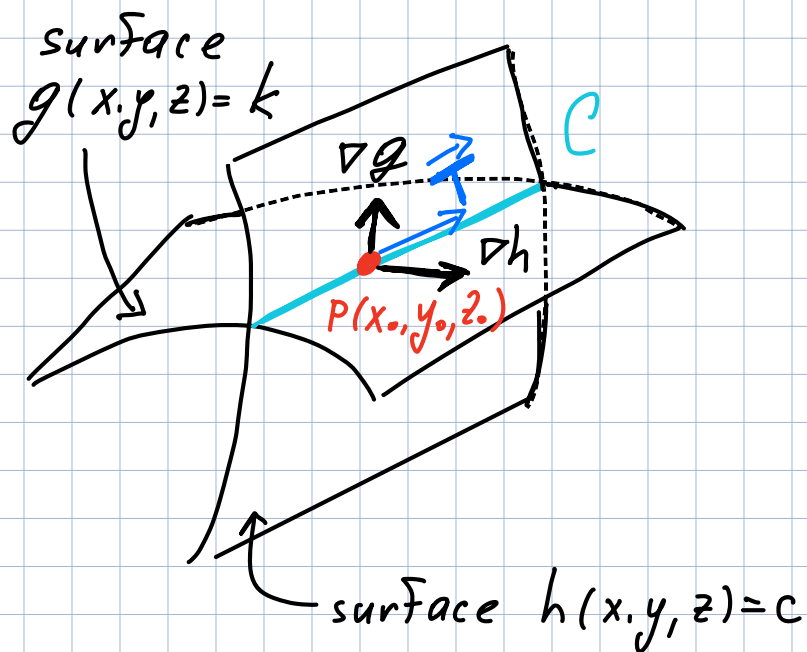
(b) evaluate f at all points found in (a)

maximum = largest value of f

minimum = smallest

Lagrange multipliers with two constraints

Problem: Find min/max values of $F(x, y, z)$ subject to two constraints: $\begin{cases} g(x, y, z) = k \\ h(x, y, z) = c \end{cases}$ $\swarrow \searrow$ define a curve C



If at $P(x_0, y_0, z_0)$ on C , F has a min or max, then directional derivative

$$D_{\vec{T}} F = 0, \quad (D_{\vec{T}} F = \nabla F \cdot \vec{T})$$

where $\vec{T}$ is the unit tangent vector along C at P .

So, $\nabla F(x_0, y_0, z_0)$ is orthogonal to $\vec{T}$

$\Rightarrow \nabla F|_P$ lies in the plane orthogonal to C at P .

This plane is defined by $\nabla g(x_0, y_0, z_0)$ and $\nabla h(x_0, y_0, z_0)$.

Therefore

$$\nabla F(x_0, y_0, z_0) = \lambda \nabla g(x_0, y_0, z_0) + \mu \nabla h(x_0, y_0, z_0)$$

$\swarrow \searrow$ Two Lagrange multipliers

Method of Lagrange multipliers

To find maximum & minimum values of $f(x, y, z)$

subject to constraints $g(x, y, z) = k$ and $h(x, y, z) = c$

① Find all values of x, y, z, λ, μ such that

$$\bullet \nabla f(x, y, z) = \lambda \nabla g(x, y, z) + \mu \nabla h(x, y, z) \leftarrow$$

$$\bullet g(x, y, z) = k$$

$$\bullet h(x, y, z) = c$$

$$\begin{aligned} f_x &= \lambda g_x + \mu h_x \\ f_y &= \lambda g_y + \mu h_y \\ f_z &= \lambda g_z + \mu h_z \end{aligned}$$

② evaluate f at all points found in (a)

maximum = largest value of f

minimum = smallest

Ex: Find the max value of $f(x, y, z) = x + 2y + 3z$ on the curve C of intersection of $\underbrace{x - y + z = 1}_{g(x, y, z)}$ (a plane) and $\underbrace{x^2 + y^2 = 1}_{h(x, y, z)}$ (a cylinder)

Sol:

$$\begin{cases} f_x = 1 = \lambda \cdot 1 + \mu \cdot 2x \\ f_y = 2 = \lambda \cdot (-1) + \mu \cdot 2y \\ f_z = 3 = \lambda \cdot 1 + \mu \cdot 0 \\ x - y + z = 1 \\ x^2 + y^2 = 1 \end{cases} \Rightarrow \lambda = 3 \text{ and } \begin{cases} 3 + 2\mu x = 1 \\ -3 + 2\mu y = 2 \end{cases} \Rightarrow \begin{cases} x = -\frac{1}{\mu} \\ y = \frac{5}{2\mu} \end{cases}$$

Now, $x^2 + y^2 = 1 \Rightarrow \left(-\frac{1}{\mu}\right)^2 + \left(\frac{5}{2\mu}\right)^2 = 1$

$$\Rightarrow \frac{29}{4\mu^2} = 1 \Rightarrow \mu = \pm \frac{\sqrt{29}}{2}$$

Solution 1: $\lambda = 3, \mu = \frac{\sqrt{29}}{2}, x = -\frac{2}{\sqrt{29}}, y = \frac{5}{\sqrt{29}}, z = 1 - x + y = 1 + \frac{7}{\sqrt{29}}$

Solution 2: $\lambda = 3, \mu = -\frac{\sqrt{29}}{2}, x = \frac{2}{\sqrt{29}}, y = -\frac{5}{\sqrt{29}}, z = 1 - x + y = 1 - \frac{7}{\sqrt{29}}$

$$f(x, y, z) = x + 2y + 3z \quad f\left(-\frac{2}{\sqrt{29}}, \frac{5}{\sqrt{29}}, 1 + \frac{7}{\sqrt{29}}\right) = \boxed{3 + \sqrt{29}} \leftarrow \begin{matrix} \text{max} \\ \text{value} \\ \text{on } C \end{matrix}$$

$$f\left(\frac{2}{\sqrt{29}}, -\frac{5}{\sqrt{29}}, 1 - \frac{7}{\sqrt{29}}\right) = 3 - \sqrt{29}$$

Ex: Plane $4x - 3y + 8z = 5$ intersects the cone $z^2 = x^2 + y^2$ in an ellipse C . Find highest and lowest points on C .

Sol: $g(x, y, z) = 4x - 3y + 8z$
 $h(x, y, z) = x^2 + y^2 - z^2$ $f(x, y, z) = z$

WANTED: max/min of f subject to $g(x, y, z) = 5$
 and $h(x, y, z) = 0$

$$\left\{ \begin{array}{l} 0 = \lambda \cdot 4 + \mu \cdot 2x \\ 0 = \lambda \cdot (-3) + \mu \cdot 2y \\ 1 = \lambda \cdot 8 + \mu \cdot (-2z) \\ 4x - 3y + 8z = 5 \\ x^2 + y^2 - z^2 = 0 \end{array} \right. \Rightarrow \left\{ \begin{array}{l} x = -\frac{4\lambda}{2\mu} \\ y = \frac{3\lambda}{2\mu} \\ z = \frac{8\lambda - 1}{2\mu} \end{array} \right\} \text{ or } \mu = 0$$

if $\mu = 0 \Rightarrow \lambda = 0$
 but then $1 \neq 8\lambda - 2\mu z$
 so, $\mu \neq 0$.

$$4x - 3y + 8z = \frac{-16\lambda - 9\lambda + 64\lambda - 8}{2\mu} = 5 \Rightarrow 39\lambda - 8 = 10\mu \Rightarrow \mu = \frac{39\lambda - 8}{10}$$

$$x^2 + y^2 - z^2 = \frac{1}{4\mu^2} (16\lambda^2 + 9\lambda^2 - 64\lambda^2 + 16\lambda - 1) = 0$$

$$\underbrace{-39\lambda^2 + 16\lambda - 1 = 0}_{-39\lambda^2 + 16\lambda - 1 = 0} \Rightarrow \lambda = \frac{16 \pm \sqrt{16^2 - 4 \cdot 39}}{2 \cdot 39} = \frac{16 \pm 10}{78}$$

$$\lambda_1 = \frac{6}{78} = \frac{1}{13} ; \quad \lambda_2 = \frac{26}{78} = \frac{1}{3}$$

Solution 1: $\lambda = \frac{1}{13}, \mu = \frac{1}{10} \left(\frac{39}{13} - 8 \right) = -\frac{1}{2}, x = -\frac{4\lambda}{2\mu} = \frac{4}{13}, y = \frac{3\lambda}{2\mu} = -\frac{3}{13}, z = \frac{2\lambda-1}{2\mu} = \frac{5}{13}$

Solution 2: $\lambda = \frac{1}{3}, \mu = \frac{1}{2}, x = -\frac{4}{3}, y = 1, z = \frac{5}{3}$

highest point on C

lowest point on C

Ex: Rectangular box without a lid is made of 12m^2 of cardboard.

What is the max possible volume?

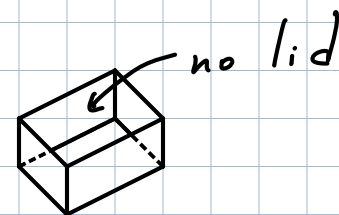
Sol: $x, y, z > 0$

$f(x, y, z) = xyz$ - volume

$g(x, y, z) = \underbrace{xy + 2xz + 2yz}_{\text{surface area}} = 12$

single constraint!

$$\begin{cases} yz = \lambda(y + 2z) & \cdot \frac{1}{yz} & 1 = \lambda\left(\frac{1}{z} + \frac{2}{y}\right) & \textcircled{1} \\ xz = \lambda(x + 2z) & \cdot \frac{1}{xz} & 1 = \lambda\left(\frac{1}{z} + \frac{2}{x}\right) & \textcircled{2} \\ xy = \lambda(2x + 2y) & \cdot \frac{1}{xy} & 1 = \lambda\left(\frac{2}{y} + \frac{2}{x}\right) & \textcircled{3} \\ xy + 2xz + 2yz = 12 \end{cases}$$



Note that $\lambda \neq 0$

$$\textcircled{1} - \textcircled{2}: z \lambda \left(\frac{1}{y} - \frac{1}{x} \right) = 0 \Rightarrow y = x$$

$$1 = \lambda \left(\frac{1}{2} + \frac{2}{y} \right) \Rightarrow 1 = \lambda \frac{4}{x} \Rightarrow \lambda = \frac{x}{4}$$

$$\textcircled{1} - \textcircled{3}: \lambda \left(\frac{1}{z} - \frac{2}{x} \right) = 0 \Rightarrow z = \frac{x}{2}$$

$$xy + 2xz + 2yz = 12 \Rightarrow 3x^2 = 12$$

$$\Rightarrow x = \pm 2$$

should be $> 0 \Rightarrow 2$

$$x = 2, y = 2, z = 1$$

$$\text{max volume: } 2 \cdot 2 \cdot 1 = \underline{4}$$